Virtualisation Basics (learn why virtualisation powers modern IT, improving efficiency and safely isolating environments).

In this room we will learn how companies have optimized there computer components to reduce costs and build flexible, scalable systems the power modern Internet through concept known as virtualization

* Have you ever considered how expensive and inefficient it would be every piece of software or every website required its own physical server? Virtualization was created to solve exactly this problem.

* your manager Asked for help on improving hardware utilization of computer that hosts a website.

lets Explore concept of virtualization and help your manager!

Learning Objectives

* Understand why managing applications on Individual physical server is inefficient
* learn how virtualization addresses hardware utilization and scalability challenges.
* understand the components of lab machine.
* learn how containers have further optimized hardware utilization for application.

Task 2 :- Virtualization overview :-

Before concept of virtualization, The rule of thumb in IT was : "One server = one application"

In early days, A business added more services, they naturally increased number of physical servers.

* This mean that if a company wanted to run website, a database, a email service etc they would have separate physical servers for each.

The problems was :-

1) High cost :- Buying multiple physical servers is expensive, not just hardware, also electricity, cooling and data center space etc.

2) low utilization :- Most Applications don't use server's full capacity. Many servers stayed at 5-20% usage, Wasting CPU, memory, storage.

3) Slow deployment :- Setting up new physical servers could take days or weeks.

4) Hard to scale :- If application suddenly needed more resources, you often had to buy another server.

In short companies were paying lot's of hardware that wasn't being fully utilized.

The Need for Sharing hardware safely and Efficiently :-

Virtualization Introduced a new idea :-

"what if multiple applications could share same physical server safely?"

A virtualization layer, called hypervisor, was introduced to act as referee b/w lab machines and allow each virtual computer to behave independently. Like a physical Computer.

The Building Analogy :-

Imagine one single person alive in entire 10 floor building.

* The person uses only one floor but must maintain entire building : electricity, cleaning water and security.

* Most of building stays empty and wasted.

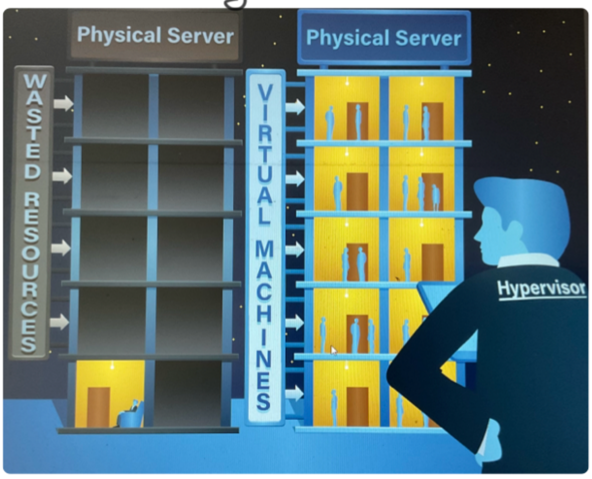
* It's expensive , insufficient and unnecessary for his needs.

Now Imagine diving building into seperate Apartments :-

* Each Apartment has its own door, walls, kitchen and privacy.

* Different people can live Independently without bothering each other.

* They all share building's main structure : electricity, water, and elevators making it cheaper and more efficient.



This is virtualization :-

* The building = The physical Server

* The Apartments = lab machines

* The tenants = applications or operating systems

* The building manager = The hypervisor (The software that divides building safely)

Each virtual computer , Known as lab Machine (VM), acts as Independent System with its own operating System , apps and Settings even though they all Share Same physical hardware underneath.

Quiz :-

1) What does virtualization enable multiple applications to share?

A) physical server

2) What is name of Software that manages resources of each lab machine?

A) hypervisor